THE SI SENTINEL

Weekday briefing on builders shaping super intelligence

TUESDAY, NOVEMBER 10, 2026

THE MORNING PICTURE

Meta goes small with Muse Ember, Anthropic ships Fable 5.2, and Musk says Grok 4.8 is built on a rewritten stack

Meta released Muse Ember, its smallest and cheapest model, and Mark Zuckerberg and Alexandr Wang pointed to independent boards that put it on the cost-per-task frontier; by evening it was on Google Cloud. Anthropic released Claude Fable 5.2, which Guillermo Rauch called "astonishingly brilliant," while Suhail Doshi objected that Anthropic deliberately held back its skill at accelerator kernel work.

Early Tuesday Elon Musk said xAI's team has dumped JAX and rewritten its training and inference stack in C and C++, that Grok 4.8 is now in reinforcement learning, and that a soft rollout to Grok Bot users comes later this week. Google added two eighth-generation Tensor Processing Unit (TPU) systems, David Heinemeier Hansson turned a pair of Nvidia DGX Sparks into Omarchy desktops, and Uncle Bob Martin took his first Cybercab ride.

I · MODELS

Meta's Muse Ember aims at the work agents repeat all day

Meta Superintelligence Labs released its smallest Muse model with a pitch about price, not peak scores. Anthropic answered at the top end with Claude Fable 5.2, and Musk gave a detailed Grok 4.8 status update.

$0.10 / $0.50	36	$0.14
Muse Ember price per million input / output tokens, below 200,000 input tokens	Ember's Artificial Analysis Intelligence Index score at max effort, level with Claude Haiku 5.5	Ember's cost per Artificial Analysis task, against $0.41 for Haiku 5.5

Mark Zuckerberg and Alexandr Wang @finkd · @alexandr_wang · Meta

Zuckerberg announced the model at 12:54pm EST:

> *"Releasing Muse Ember today -- our smallest model that delivers higher intelligence per dollar than others in its class. Built for token-heavy work that agents repeat all day: tool calls, coding subtasks, routing, extraction, and long document parsing. Available today on the Meta Model API and in Muse Code."*
>
> MARK ZUCKERBERG · @FINKD · 12:54PM EST · POST

Wang, who runs Meta Superintelligence Labs, added that Ember "offers nearly Muse Spark 1.3 (max) intelligence at a 17x cheaper price point" and "is the first model we've released trained using Muse Wonder Full," the lab's largest model. Meta's launch post gives Ember a one-million-token context window and native text, image, audio and video input; as Wang's thread put it, "wonder can plan, ember can handle the volume."

Most of Wang's afternoon went to quote-posting outside scorekeepers. Artificial Analysis scored Ember 36 on its Intelligence Index, matching Claude Haiku 5.5 at a third of the cost per task, while noting that knowledge is its main weakness: it declines to answer 64% of its factual questions, which keeps its hallucination rate at 17.4%, half of Haiku's. Wang's summary was "matching haiku at 3x cheaper cost per task, and smarter than luna." Vals AI called it the cheapest model on its index at $0.15 per test, and the Code Arena clocked its generated websites at about $0.04 each. "Back in my day, these would cost you O($10,000) in custom dev shop costs," Wang wrote. Box showed Ember finding a $2.48 million misallocation across 13 documents in a private-fund capital

call, and Wang confirmed that "crouton-alpha," a stealth model on OpenRouter, "was an early version of muse ember."

By late afternoon Zuckerberg posted that "Muse Ember, Muse Spark 1.3, and Muse Wonder are now available on Google Cloud." Sundar Pichai replied at 5:41pm EST: "Congrats and welcome to the model garden, excited to serve our Google Cloud customers!"

Anthropic's Claude Fable 5.2 @rauchg · @levie · @Suhail

Anthropic released Claude Fable 5.2 and its restricted twin, Claude Mythos 5.2, the same model with fewer safeguards for verified cybersecurity and life-sciences groups. It says the pair work up to 30% faster than the 5.1 models at $8 per million input tokens and $40 per million output tokens, and that in testing they designed three working electronic devices and formalized the proof of Fermat's Last Theorem in 30 hours. Guillermo Rauch, chief executive of Vercel, said his company tested it before release, at 2:42pm EST:

> *"Best way to summarize it: astonishingly brilliant. One interesting takeaway: it's very clear that this new generation of models can take on unprecedented bodies of parallel work. To actually seize this intelligence, you need cloud sandboxes. Full stop. The model can delegate and orchestrate far more than a single computer can handle."*
>
> GUILLERMO RAUCH · @RAUCHG · 2:42PM EST · POST

Aaron Levie, chief executive of Box, quoted Anthropic's showcase of pre-release projects and predicted "autonomous collaborators in a number of industries that haven't yet had their claude code or codex moment," naming animation, film production, gaming, manufacturing, architecture, product design and robotics.

Suhail Doshi, founder of Playground, took the other side. Anthropic's announcement says that during training it "took steps to limit advances" in capabilities that could most directly accelerate AI development, so on accelerator kernel optimization the model performs at about the level of Claude Opus 5.5. Doshi wrote: "The capitalist in me gets it but this won't go unanswered by society. Anthropic will continue to gatekeep until your products can't use their models or you can't make certain things that they'll compete with you on." Told that only biology and model training had been restricted before, he answered: "Now GPU kernels."

Elon Musk · Grok 4.8 @elonmusk · CEO, SpaceX and Tesla

Asked Monday evening when to expect Grok 4.8, Musk answered early Tuesday, at 4:35am EST:

> *"We have dumped JAX/SGLang and rewritten the entire RL and inference stack from scratch in C/C++, which took some extra time. Training (aka pre-training) is done. Lot of cleanup was needed on the SFT data to get that mostly right. Now, 4.8 is in RL and the evals are starting to climb. We will do a soft rollout to Bot users later this week and see how it goes. Grok 4.8 will bear no resemblance to 4.7 (thankfully). It is not yet incredible, but, this time, the foundations are built right, so I expect it to evolve to be something special."*
>
> ELON MUSK · @ELONMUSK · 4:35AM EST · POST

RL is reinforcement learning, the post-training stage; SFT is supervised fine-tuning. Asked why the team left Rust behind for C and C++, he replied: "Real men manage their own memory." In a thread with the developer account @tetsuoai he described the new code as "C with a little C++, but not too much," added "At SpaceX, we write our own headers and calculate our own time (includes leap seconds)," credited the new Grok training and inference stack to "amazing engineers," and noted "even some assembly." "One day, super intelligence will write a better software stack, but that day has not yet come," he wrote. Asked how 4.8 would compare with Anthropic's Claude Opus 5.5, he said: "Not as good. Opus 5.5 is a masterpiece. But Grok 5 should exceed Opus 5.5 and maybe that happens in 2 or 3 months, with continued acceleration thereafter." At 5:34am EST he added that "the Starlink software, built for managing the movement of vast amounts of data through space and to the ground, is also very well suited to training super intelligence."

II · CHIPS AND MACHINES

Google ships four TPU systems in one generation

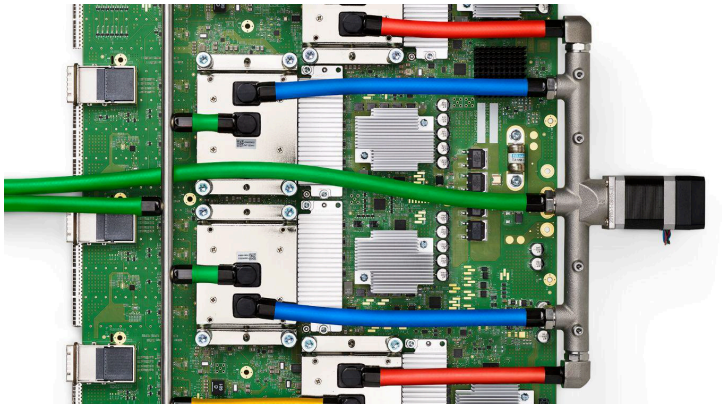
Google added a confidential-computing TPU and a dual-chip TPU for giant context windows, its third and fourth eighth-generation systems this year. Musk defended Tesla's reported memory cut for its next robot chip, and an open-source Linux distribution moved onto Nvidia's desk-sized AI computer.

Sundar Pichai and Lisa Su · TPU 8tx and 8i Max

@sundarpichai · @LisaSu

Amin Vahdat, Google's chief technologist for AI and infrastructure, announced the two systems on Monday morning. In Google's blog post, TPU 8tx uses the same silicon as the training-focused TPU 8t, paired with fifth-generation AMD EPYC host processors, and keeps data and model weights protected "even while in active use," for regulated companies and governments; Google claims up to 2x lower latency than TPU 8t. TPU 8i Max puts two inference-focused TPU 8i chips in one package, doubling compute, on-chip memory (SRAM) and high-bandwidth memory (HBM) bandwidth, and fits twice the

silicon in the same rack. Vahdat also listed TorchTPU, which runs native PyTorch and Hugging Face models on TPUs, and an open-source TPU Sync library that he said cuts KV-cache transfer latency by more than 90%.



A TPU v4 board from an earlier Google generation, with four liquid-cooled chip packages, as published in Google's 2023 TPU v4 paper.

IMAGE: NORMAN P. JOUPPI ET AL., "TPU V4" / WIKIMEDIA COMMONS · CC BY 4.0 (CROPPED)

Pichai quoted the announcement at 1:19pm EST:

> *"Four distinct TPU systems in a single generation in one year! Instead of waiting years for a traditional semiconductor cycle, hardware engineers co-designing systems alongside Google Deepmind means we can keep innovating quickly for customers."*
>
> SUNDAR PICHAI · @SUNDARPICHAI · 1:19PM EST · POST

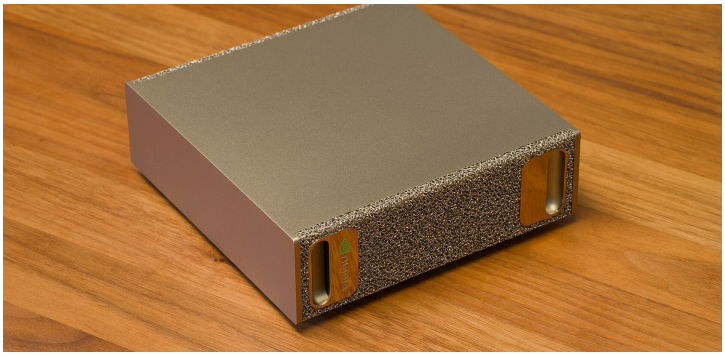
AMD's chief executive, Lisa Su, wrote: "We're proud to partner with you, with @AMD EPYC serving as the host processor for TPU 8tx."

Elon Musk · Tesla's AI6 memory @elonmusk

Tesla investor Herbert Ong shared a China Flash Market report that Tesla has cut the low-power LPDDR6 memory on its next-generation AI6 chip for the Optimus robot from 216 GB to 144 GB, a third less, which he read as Tesla designing around a memory shortage caused by AI servers. Musk did not dispute the figure, but replied at 10:20am EST: "Memory bandwidth was held constant though." The distinction matters for a robot's real-time control loop, which depends more on how fast data moves than on how much is stored.

David Heinemeier Hansson · Omarchy on the DGX Spark @dhh · creator of Ruby on Rails and Omarchy

Heinemeier Hansson thanked Nvidia for a pair of DGX Spark desktop AI computers at 9:25am EST: "Not only are the machines amazing, but they look absolutely incredible too. Omarchy is fully up and running now! Working to tune the out-of-box local AI offerings. All shipping in 4.6 shortly." He replaced Nvidia's operating system on one unit, which is now his main desktop, and left the second running headless. For a single Spark, he said, "Qwen3.8 Flash Next is the best bet," running at about "60tps", and he plans to make it "even easier to use inference from any machine on your network on any other machine."



An Nvidia DGX Spark, the desktop AI computer now running Omarchy on Heinemeier Hansson's desk, photographed in May 2026.

PHOTO: DANIEL LU (USER:DLLU) / WIKIMEDIA COMMONS · CC BY-SA 4.0

He also wrote that the first Snapdragon X1 laptops "are still very capable and shockingly energy efficient," with "stellar bargains to be had on these devices because Windows flopped so hard with the initial rollout"; on an Asus ZenBook A14, he said, "Everything except deep sleep is working." He shared a user running "macOS running inside Omarchy as a VM on a MacBook," and a 2008 ThinkPad X200 that installed Omarchy 4.5 in two and a half minutes.

III · ROBOTAXIS, FACTORIES AND ROCKETS

A first Cybercab ride, and the robotaxi bet

A software veteran took his first ride in Tesla's two-seat robotaxi as Palmer Luckey and Musk argued over what the business is worth. Tesla broke ground on a $10 billion solar plant, and SpaceX said Grok now beats its human inspectors.

Uncle Bob Martin, Palmer Luckey and Elon Musk

@unclebobmartin · @PalmerLuckey · @elonmusk

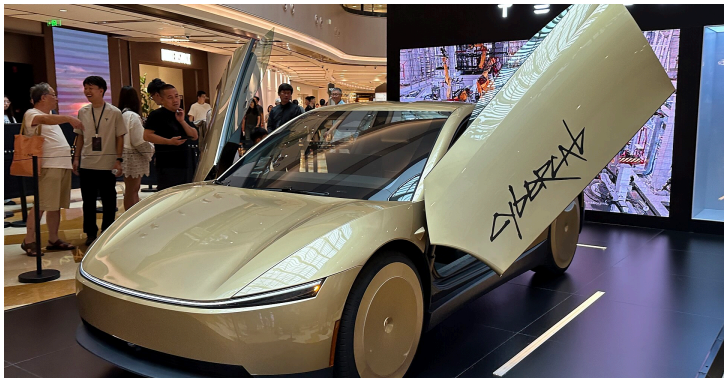
Robert "Uncle Bob" Martin, author of *Clean Code*, posted at 2:08pm EST: "Our first cybercab ride. Fantastic!"

On X, Carl Hancock argued that turning Tesla into a taxi company made little sense, since the five largest ride-hailing companies have about 16 million drivers and a combined market value of $175 billion. Luckey, the founder of Anduril, replied at 12:35am EST:

> *"Take that ridesharing revenue ($150B+) and subtract the cost of paying 16,000,000 drivers. That is the bet."*
>
> PALMER LUCKEY · @PALMERLUCKEY · 12:35AM EST · POST

Musk went further at 4:44am EST: "Cybercab will do a majority of all passenger miles in any jurisdiction where it is not

prohibited by law."



A Tesla Cybercab, the two-seat robotaxi with no steering wheel or pedals, on display in a shopping mall, Sept. 19, 2026.

PHOTO: JUSTANOTHERCARDESIGNER / WIKIMEDIA COMMONS · CC0

Tesla's Project Crystal Sun @elonmusk

Musk quoted the news that Tesla has started construction on its $10 billion solar manufacturing plant near Houston with three words: "Project Crystal Sun." As Teslarati reported, site work has begun in Fort Bend County, with construction planned through 2028 and commercial operations targeted for the first quarter of 2029. The project filing summarized by Sawyer Merritt describes a vertically integrated plant, from ingot and wafer manufacturing through cell coating, metallization and testing.

SpaceX · Grok on the factory floor @elonmusk

At the UP.Summit conference in Florida last week, SpaceX's vice president of launch, Kiko Dontchev, said Grok has largely automated the search for foreign object debris (FOD), the stray fragments that can wreck an engine, Ars Technica reported: "We'll have an inspector go in and borescope a tube, and we find that humans are about 94 percent perfect (at) finding a piece of it. Grok analyzing the same video is 99.9 percent perfect." He called SpaceX "fundamentally an infrastructure company at this point." Musk quoted the clip with one word: "Grok."

Dontchev also said Falcon 9 is on the road to retirement because its second stage is still thrown away on every flight, while Starship is meant to catch both stages. "Pad 39A is effectively complete," he said of Starship's first Florida launch tower at Kennedy Space Center, adding that SpaceX hopes to barge the first boosters and ships to the Cape in December. A Florida Starship, he said, could carry up to 20 times the Starlink bandwidth of a single Falcon 9.

> QUICK HIT
>
> **Kevin Weil** cheered "Two of my favorites, coming together!" as Stoke Space and Planet signed a multi-launch deal for Stoke's reusable Nova rocket, starting with a pathfinder flight in 2027.

IV · AGENTS AT WORK

Rauch wants every system on one protocol

Vercel's MCP servers can now call agents back when something happens. Kanjun Qiu pitched Imbue Studio as a computer you own, François Chollet liked a company run by agents, and Greg Brockman pointed to a restaurant number.

Guillermo Rauch · MCP Events @rauchg · CEO, Vercel

Vercel announced that its Model Context Protocol (MCP) server now publishes events, so an agent can subscribe to deployment successes or errors and get a signed webhook instead of polling. Rauch quoted it at 3:33pm EST:

> *"So bullish on MCP. Its utility exceeds the "M" part: even AI aside, it'll be a civilization-scale accomplishment to get all our systems communicating through one universal open protocol. A missing piece was how to get arbitrary callbacks across sessions. MCP could read and write, but it couldn't tell you later "this event happened." MCP Events fixes this."*
>
> GUILLERMO RAUCH · @RAUCHG · 3:33PM EST · POST

Vercel is adding support in its own MCP server, he wrote, so an agent that starts a deployment "can subscribe to deployment.succeeded" instead of polling for the result.

Kanjun Qiu · Imbue Studio @kanjun · CEO, Imbue

Programmer Steve Krouse wrote that he has "always dreamed of a cloud-first operating system, where my phone and computer were just thin windows into it." Qiu replied that Imbue Studio, launched last month, "is meant to be this," cloud-first and open source. When Krouse said he wanted something lower-level, she explained that "Studio runs a normal Linux VM with agents," that users can "change anything you want in the VM," and that "you can literally change out the OS to Omarchy if you really wanted." Its mission, she wrote, "is to give control of digital surfaces back to the user (vs having people's attention be constantly preyed on)." Asked about the Silicon Valley fixation on agency, she said she now prefers "authorship" and "sovereignty," because "agency has been co-opted to mean something that feels very controll-y."

François Chollet · autonomous businesses @fchollet · co-founder, Ndea

Start-up founder Sam Pasupalak introduced "Steve," billed as an AI chief executive that runs three acquired businesses, plus a $2 million program to buy five more. Chollet wrote at 12:23pm EST: "Acquiring distressed or stagnant SaaS apps and handing the keys to an agent harness is the ultimate real-world

eval. Just P&L as the objective function and the market as the feedback signal." SaaS is software as a service.

Greg Brockman · agentic commerce @gdb · president, OpenAI

Brockman wrote "integrating with OpenAI can drive a lot of business," quoting e-commerce analyst Juozas Kaziukėnas, who reported that OpenTable's native integration with OpenAI "drove 190,000 people to restaurants in just the month of September."

Uncle Bob Martin · clean code for agents @unclebobmartin

Answering a critic who paraphrased him as saying clean code no longer matters, Martin wrote: "The principles of clean code are as important, if not more important, as always. Agents get just as confused by messy code as humans do." He added: "I do not 'vibe code'. All the same old engineering principles apply to what I do. The only personal behavior I have changed is to disengage from syntax." To a reader who argued that fully trusted models should simply emit machine code, he replied: "AIs use languages the same way we do. They state what they want in a high level syntax and then let a program do the dirty work of turning it into binary."

Fei-Fei Li · World Labs Atlas @drfeifei · co-founder, World Labs

World Labs said its Atlas model ranks first in Rapidata's camera-movement benchmark, which compares 19 world and video models across 600 prompts and more than 513,000 human judgments. Li wrote: "Camera control is so critical for many creative and design work."

Kanjun Qiu · platform lock-in @kanjun

Qiu also turned on X itself over a new on-screen hand animation, at 1:05am EST: "This is a classic example of platform lock-in allowing the platform to increasingly antagonize the user to juice its metrics because it verticalizes through the UI layer. It makes me want to start an X scraper community that puts all X posts on a public chain, using Bluesky's protocol, to free all of us from this grossness." When a reader urged her to build it, she replied: "I do feel like in the age of agents, this battle is going to be fought, and relatively soon."

QUICK HIT

Mustafa Suleyman, chief executive of Microsoft AI, will debate "AI promises abundance. How realistic is that?" with Sarah Guo and Aaron Bastani at the Commonwealth Club in San Francisco on Nov. 18. "We probably won't agree. That's the point."

ALSO ON THE WIRE

Patrick Collison congratulated Brian Chesky on his $100 million gift to San Francisco parks, the city's largest from a single donor; Suhail Doshi called it "City of the future!" Tim Sweeney suggested that Apple developers still lobby to get rivals' apps removed, after one developer lost a six-year account. Guillaume Lample joked that Mistral's model "looked at the prisoner's dilemma and saw the Dark Forest," about a game in which AI models run countries with a nuclear option. Jim Fan streamed his keynote at the Conference on Robot Learning (CoRL) in Austin.

Takeaways

1. The model race now runs on cost per task as much as capability: Meta's Ember matches Haiku 5.5's score at about a third of the price.
2. Anthropic is shipping its strongest public model while deliberately holding back its skill at AI research work such as accelerator kernels, and builders are noticing.
3. xAI is betting on its own low-level stack: Grok 4.8 runs on rewritten C and C++, with a soft rollout this week and Grok 5 aimed at Opus 5.5.
4. Custom silicon cycles are shrinking: Google brought out four TPU systems in one generation in a single year.
5. The robotaxi case is simple arithmetic to its backers: ride-hail revenue minus 16 million drivers' pay.
6. Agents are getting plumbing, with MCP events, owner-controlled computers and businesses run end to end by agents.

On the beat

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Elon Musk, Uncle Bob Martin, Patrick Collison, Palmer Luckey, Tobi Lütke, David Heinemeier Hansson, François Chollet, Fei-Fei Li, Greg Brockman, Mustafa Suleyman, Paul Graham, Jim Fan, Alexandr Wang, Mark Zuckerberg, Lisa Su, Sundar Pichai, Kanjun Qiu, Guillermo Rauch, Guillaume Lample, Michael Dell, Aaron Levie, Tim Sweeney, Kevin Weil, Suhail Doshi.

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Marc Andreessen, Jensen Huang, Ashok Elluswamy, Andrej Karpathy, Sam Altman, Demis Hassabis, Dario Amodei, John Carmack, Yann LeCun, Satya Nadella, Vitalik Buterin, Jeff Dean, Andrew Ng, Michael Truell, Tom Brown, Clément Delangue, Mira Murati, Noam Brown, David Singleton, Ilya Sutskever, Joe Gebbia, Nat Friedman, Arthur Mensch, Mark Chen, Aidan Gomez, Albert Gu, Richard Sutton.

Added Amin Vahdat (@AminVahdat), Google's chief technologist and senior vice president for AI and infrastructure, who leads its TPU and data-center systems work. The list now stands at 52.

The SI Sentinel is compiled from public posts on X and published reporting. Quotes are reproduced verbatim unless marked as translated or paraphrased; emoji in quoted posts are omitted.